

MODULE 11 · DOMAIN 4 - DESIGN COST-OPTIMIZED ARCHITECTURES

Storage Cost and Lifecycle

S3 classes, tiering and getting rid of what you do not need.

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You pay for **storage**, for **retrieval**, and for **how quickly** you want it back. Cheaper storage means slower or costlier retrieval.

The S3 storage classes

CLASS	USE IT FOR	RETRIEVAL
S3 Standard	Frequently accessed data	Immediate
S3 Intelligent-Tiering	Unknown or changing access patterns	Immediate, no retrieval fee
S3 Standard-IA	Infrequent access, needed immediately	Immediate, retrieval charge
S3 One Zone-IA	Infrequent, and re-creatable if the AZ is lost	Immediate, retrieval charge
Glacier Instant Retrieval	Archive accessed a few times a year	Milliseconds
Glacier Flexible Retrieval	Archive, occasional access	Minutes to hours
Glacier Deep Archive	Long-term retention, rarely read	Twelve hours or more

ON THE EXAM

Minimum durations apply: 30 days for the IA classes, 90 for Flexible Retrieval, 180 for Deep Archive.

Intelligent-Tiering or lifecycle policies?

INTELLIGENT-TIERING

When you do not know the pattern

- Moves objects automatically on real access
- Small monitoring charge per object
- No retrieval fees between tiers
- Nothing to design or maintain

LIFECYCLE POLICY

When the pattern is known by age

- Transition at fixed ages you specify
- No monitoring charge
- You choose exactly which classes
- Can expire objects entirely

THE TIEBREAKER

'Access patterns are unknown or unpredictable' means Intelligent-Tiering. 'Data is accessed for thirty days then rarely' means a lifecycle policy.

Other storage cost levers

- ◆ **Lifecycle rules can expire objects** and clean up incomplete multipart uploads, which quietly accumulate cost.
- ◆ **S3 Storage Lens** gives visibility of usage and cost across buckets and accounts. Use it to find what to fix.
- ◆ **Delete old EBS snapshots.** They persist after the volume is gone and are commonly forgotten.
- ◆ **gp3 is usually cheaper than gp2** for the same performance, and throughput is configured separately.
- ◆ **Unattached EBS volumes** bill at full rate. So do idle load balancers and unassociated Elastic IPs.

ON THE EXAM

Availability warning: S3 Select and S3 Glacier Select closed to new customers in July 2024. Use Athena for querying data in place.

A company stores application logs in S3 Standard. Logs are queried heavily for the first thirty days, occasionally for the following sixty days, and must then be retained for seven years for compliance but are almost never read.

The bucket holds four hundred terabytes and grows by twenty terabytes a month.

Which approach is MOST cost-effective?

- ☐ A A lifecycle policy to Standard-IA at 30 days and Glacier Deep Archive at 90 days
- ☐ B Enable S3 Intelligent-Tiering on the bucket
- ☐ C A lifecycle policy moving everything to Glacier Deep Archive at 30 days
- ☐ D Keep everything in S3 Standard and compress the logs

◆ PAUSE AND DECIDE

Which approach is MOST cost-effective?

- A

 IA at 30 days, Deep Archive at 90
Matches the described pattern exactly: hot for 30 days, occasional for 60, then seven years of cheap retention.
- B

 Intelligent-Tiering
Sensible when the pattern is unknown, but here it is described precisely. You would pay a monitoring fee to discover what you already know.
- C

 Deep Archive at 30 days
Cheapest storage, but the days 30 to 90 window needs occasional access, and Deep Archive retrieval takes twelve hours.
- D

 Compress and keep in Standard
Reduces volume but keeps seven years of rarely-read data in the most expensive class.

THE SIGNAL

When the question describes the access pattern by age, encode it in a lifecycle policy. Intelligent-Tiering is for when it does not.

A media company stores user-uploaded content in S3. Some files go viral and are accessed constantly for months; most are never opened again after the first week. There is no way to predict which will be which.

The company wants to reduce storage cost without risking slow access to any file.

Which storage class should be used?

- ☐ A S3 Standard-IA with a lifecycle policy at seven days
- ☐ B S3 Intelligent-Tiering
- ☐ C S3 One Zone-IA
- ☐ D S3 Glacier Instant Retrieval

◆ PAUSE AND DECIDE

Which storage class should be used?

- A** Standard-IA at seven days
Would push viral files into IA and then charge retrieval on every one of their many reads. Potentially more expensive than Standard.
- C** One Zone-IA
User-uploaded content cannot be re-created if the Availability Zone is lost, so single-zone durability is inappropriate.

- B** Intelligent-Tiering
Unpredictable per-object access is exactly its purpose. Viral files stay in the frequent tier, forgotten ones move down, and there are no retrieval fees.
- D** Glacier Instant Retrieval
Retrieval is fast but charged per request, which is punishing for files being read constantly.

THE SIGNAL

This is the deliberate inverse of scenario one. Unpredictable per-object access is the Intelligent-Tiering signature.

A company must retain medical imaging records for twenty-five years to satisfy regulation. In practice fewer than one percent are ever retrieved, and when a retrieval is needed a wait of up to a day is acceptable.

The archive is nine hundred terabytes and the priority is the lowest possible storage cost.

Which storage class should be used?

- ☐ A S3 Glacier Deep Archive
- ☐ B S3 Glacier Flexible Retrieval
- ☐ C S3 Standard-IA
- ☐ D S3 One Zone-IA

◆ PAUSE AND DECIDE

Which storage class should be used?

- A

Glacier Deep Archive

The cheapest storage AWS offers, and its twelve-hour retrieval sits comfortably inside the stated one-day tolerance.
- B

Glacier Flexible Retrieval

Faster retrieval than needed, at a higher storage price. The question did not ask for minutes.
- C

Standard-IA

Immediate access the requirement does not need, at several times the storage cost over twenty-five years.
- D

One Zone-IA

Cheaper than Standard-IA but far dearer than Deep Archive, and single-zone durability is wrong for irreplaceable medical records.

THE SIGNAL

Match retrieval speed to what is actually required. Paying for faster retrieval than the question asks for is a wrong answer.

A company generates intermediate processing files in S3. They are read by a downstream job within hours and are never needed afterwards, but nobody has ever deleted them. The bucket now holds two hundred terabytes of these files.

The files can be regenerated from source data at any time if they were somehow lost.

Which combination is MOST cost-effective? (Choose two.)

- ☐ A Store the intermediate files in S3 One Zone-IA
- ☐ B Add a lifecycle rule expiring the files after seven days
- ☐ C Move the files to Glacier Deep Archive after one day
- ☐ D Enable versioning so old files can be recovered

◆ PAUSE AND DECIDE

Which combination is MOST cost-effective? (Choose two.)

- A

 One Zone-IA
Re-creatable data is precisely the case One Zone-IA exists for, and it costs about twenty percent less than Standard-IA.
- B

 Expire after seven days
The data is never needed after a few hours. Deleting it removes the cost entirely rather than reducing it.
- C

 Deep Archive after one day
Archiving data that is never read again pays to keep it forever, plus a 180-day minimum charge. Deletion is cheaper than any archive.
- D

 Enable versioning
Keeps additional copies of re-creatable data, which increases cost while solving nothing.

THE SIGNAL

The cheapest storage class is no storage. If data is genuinely not needed, expiry beats archiving every time.

A finance team reports the S3 bill has risen steadily for two years across sixty buckets in fourteen accounts. Nobody can explain the increase, and there is no inventory of what each bucket contains or how it is accessed.

The team wants to find out where the cost is concentrated before making changes.

What should the solutions architect recommend?

- ☐ A Enable S3 Storage Lens with the advanced tier across the organisation
- ☐ B Apply Intelligent-Tiering to every bucket immediately
- ☐ C Set a lifecycle policy on every bucket moving objects to Glacier at 30 days
- ☐ D Enable S3 Versioning to track object changes over time

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

S3 Storage Lens
Gives organisation-wide visibility of storage, growth and access patterns, which is what the team actually asked for.
- B

Intelligent-Tiering everywhere
A defensible later step, but applying it blindly adds monitoring charges on buckets that may simply need deleting.
- C

Blanket Glacier lifecycle
Would break any bucket needing immediate access, and archives data that might be better expired.
- D

Enable versioning
Keeps every previous version, which increases storage cost. It answers a different question entirely.

THE SIGNAL

The question asks how to find out, not what to change. Measure before acting, exactly as with Compute Optimizer in module ten.

An account shows unexpected EBS charges. Investigation finds one hundred and forty unattached volumes left behind by terminated instances, three thousand snapshots dating back four years, and twelve Elastic IP addresses not associated with any instance. The team wants to reduce cost and prevent this recurring.

Which combination addresses this? (Choose two.)

- ☐ A Delete the unattached volumes and unassociated Elastic IPs, and apply a Data Lifecycle Manager policy to expire old snapshots
- ☐ B Convert the unattached volumes from gp2 to gp3
- ☐ C Use AWS Trusted Advisor cost optimisation checks to find idle resources on an ongoing basis
- ☐ D Purchase Reserved Instances to offset the EBS charges

◆ PAUSE AND DECIDE

Which combination addresses this? (Choose two.)

- A

 Delete the waste, automate snapshot expiry
Removes the current cost and stops snapshots accumulating again through an automated retention policy.
- B

 gp2 to gp3 conversion
A real saving on volumes you are using. These volumes are attached to nothing, so the correct action is deletion.
- C

 Trusted Advisor cost checks
Surfaces idle volumes, unassociated addresses and low-utilisation resources continuously, which addresses the recurrence.
- D

 Reserved Instances
Reservations apply to compute, not to EBS storage or Elastic IP charges.

THE SIGNAL

Two requirements, two answers: fix the current waste, and put something in place so it does not return.

Module 11 takeaways

- ◆ **Pattern known by age** means a lifecycle policy. **Pattern unknown** means Intelligent-Tiering.
- ◆ **Match retrieval speed to the requirement.** Paying for faster than asked is a wrong answer.
- ◆ **One Zone-IA** only when the data can be re-created.
- ◆ **If it is never needed again, expire it.** Archiving costs more than deleting.
- ◆ **Unattached volumes, old snapshots and idle Elastic IPs** bill silently. Trusted Advisor finds them.

ON THE EXAM

Next: network cost, which is where most surprise charges on an AWS bill actually come from.